Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

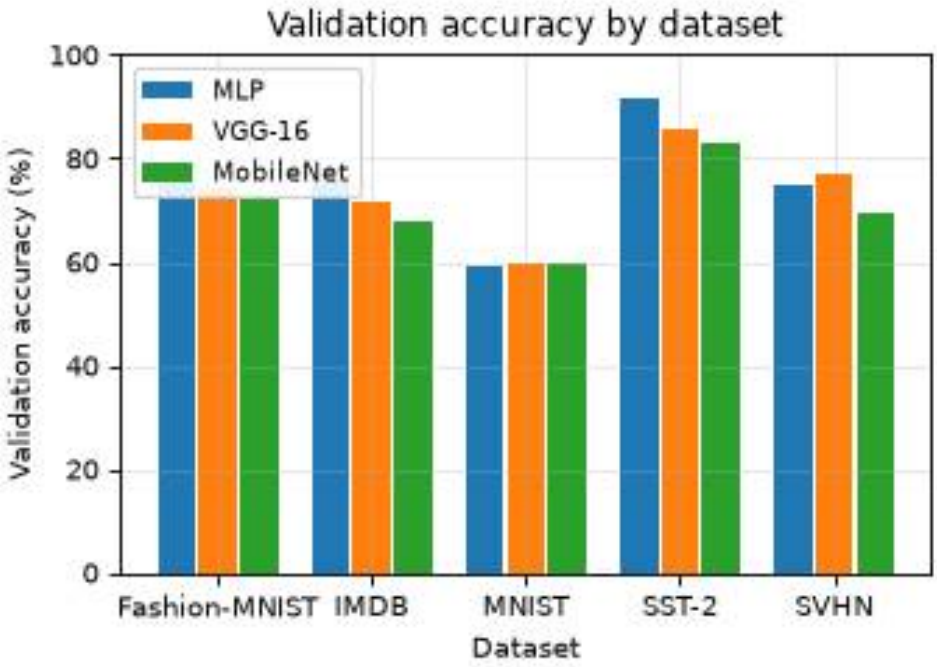


Figure 1: Validation accuracy by dataset for the models in this experiment.

On MNIST, MobileNet obtains a validation accuracy of 60.0% (Figure 1). MLP outperforms VGG-16 by 16.1 percentage points on SST-2. VGG-16 reaches 67.6% validation accuracy on IMDB (Figure 1).

Figure 2 shows the accuracy by dataset.

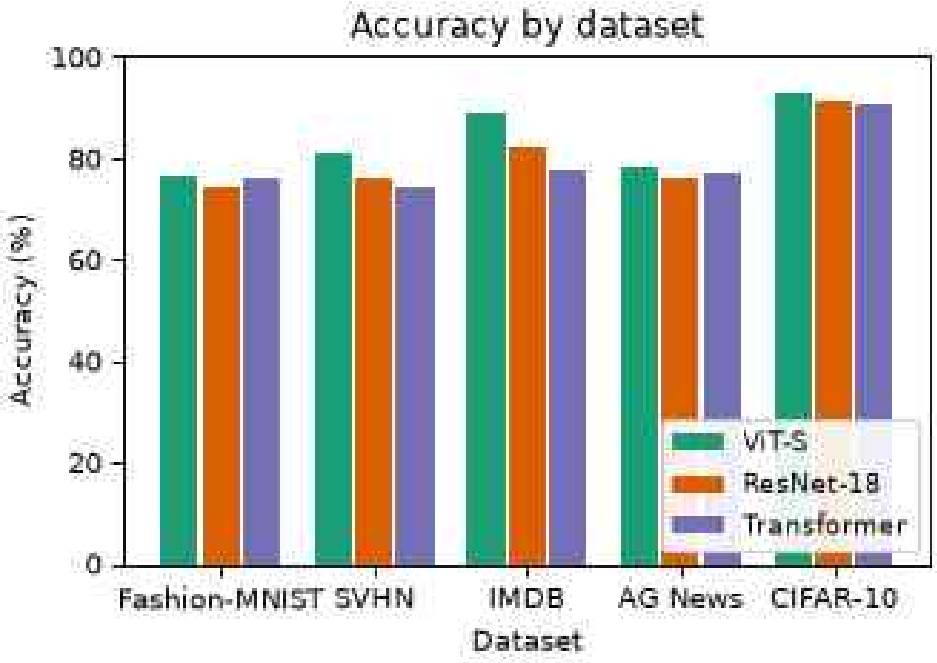


Figure 2: Accuracy by dataset for the models in this experiment.

ViT-S reaches 88.8% accuracy on IMDB. ViT-S achieves 88.0% accuracy on CIFAR-10. On Fashion-MNIST, Transformer obtains a accuracy of 26.0%.

3 Discussion

Random Forest reaches 82.2% accuracy on Tiny-ImageNet. The differences between the models are discussed in the next section.